A Systematic Literature Review of Audio Signal Processing Methods for Infant Cry Recognition and Interpretation

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***Abstract*— This study reviews audio feature extraction methods for infant cry analysis, crucial for early neonatal diagnosis. It examines 28 articles from 2020 to 2024, selected through a structured search across multiple digital libraries. While methods like MFCCs are widely used, combining techniques improves accuracy in cry classification. The review highlights the need for more robust, noise-resistant approaches and calls for further research to enhance cry recognition systems for better infant care.**

Keywords—infant cry, baby cry, audio signal processing, audio signal, audio extraction features

I. INTRODUCTION

Baby crying is a key form of communication, especially in the neonatal stage, signaling pain, hunger, or discomfort. In clinical settings like NICUs, monitoring cries can indicate a baby's health. To support this surveillance, automated systems based on speech recognition using several techniques such as Mel-Frequency Cepstral Coefficients (MFCC) have been developed as part of an effort to detect specific types of cries [1][2]. Audio signal processing methods play a critical role in analyzing cry signals, extracting meaningful features, and developing algorithms that can differentiate between various types of cries, such as hunger, pain, or discomfort [3][4][5][6][7][8]. This systematic literature review aims to comprehensively analyze the existing audio signal processing techniques used in infant cry recognition, highlighting their effectiveness, limitations, and potential areas for improvement [9][10][11][12][13]. By integrating current research findings, this review seeks to provide a foundation for future advancements in this field, ultimately contributing to better caregiving and early medical intervention for infants.

A. Related Survey

The growing interest in infant cry recognition and interpretation has led to the development of various audio signal processing methods designed to analyze and classify infant cries effectively. Previous surveys and reviews have explored multiple aspects of this field, including the challenges of cry signal acquisition, feature extraction techniques, and classification algorithms [14], [15][16][17][18][19][20]. In this section, we present a summary of related surveys that have enhanced the understanding of audio signal processing techniques for recognizing infant cries and interpretation [21][22][23][24]. By examining these surveys, we aim to identify gaps in the literature and establish a foundation for a more holistic review of the state-of-the-art techniques in this domain.

- Hua-Nong et al. This study examines the use of hybrid features, such as MFCC, Chromagram, Mel-scaled Spectrogram, Spectral Contrast, and Tonnetz, in conjunction with deep learning models to classify asphyxia cries[25]. The study is restricted to the classification of asphyxia cries and does not explore whether the same hybrid features and deep learning models can be effectively utilized to identify other types of infant cries, such as those triggered by hunger, pain, or discomfort.
- Gianpaolo et al. This document outlines an automated, self-training system for detecting infant cries, specifically designed to aid neonatologists in identifying cries amidst noisy hospital settings. The system aims to support early diagnosis and analysis of neonatal conditions[26]. This study shows that the features proposed to be extracted may not be sufficient to distinguish infant cries from all types of noise and sounds in the neonatal care environment.
- Gabriele et al. Explaining that the minor second interval, which is usually associated with sadness in music across cultures, may stem from its prevalence in infant crying, indicating the possibility of a shared emotional language between music and vocal expressions[27]. This research requires automated feature extraction methods to achieve accurate results.
- Khosro et al. This article presents a DeepSVM-based system to classify infant cries with high accuracy, using

fractal descriptors and repeated neighbourhood component analysis for feature extraction and selection, which can improve infant health monitoring in the NICU and home[28]. Future research should investigate additional cases.

- Lukas et al. The study describes the InfaSafe System, developed by researchers at the Illinois Institute of Technology, is a comprehensive non-invasive infant monitoring platform that utilises AI and advanced computing to provide real-time insights into infant health, with the primary goal of reducing Sudden Unexpected Infant Death (SUID) through advanced pose estimation, respiratory surveillance, and cry analysis [29]. The audio extraction feature used in this research can be developed for other cases.

This research reviews audio processing techniques for recognizing infant cries, assessing their strengths and weaknesses. It helps researchers choose suitable methods and highlights research gaps for future improvement. Advances in this field aim to improve infant care and enable early medical intervention.

II. METHODS

This chapter details the systematic search strategy used, including the criteria for selecting relevant studies and the keywords used. It also describes the data extraction process, the techniques used to evaluate the quality and relevance of each study, and the methods for synthesising the findings.

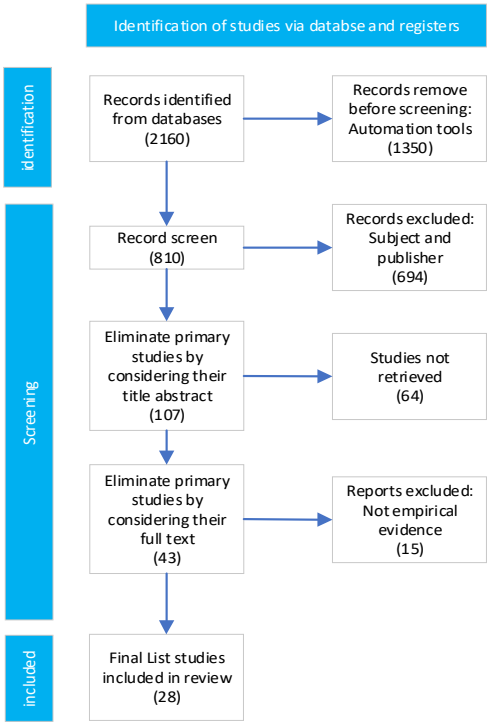


Fig. 1. Selection Literature Flowchart
(Source: Data Processed by the Author 2024, Yogyakarta)

Figure 1 shows the process of identifying and selecting relevant studies for a survey. It begins by setting research questions and inclusion/exclusion criteria, followed by searching databases with predefined terms. Results are filtered by subject and publication year, and titles and abstracts are reviewed. Full texts are then examined, and studies not meeting the criteria are excluded. Citation chaining and manual searches may also be used to find additional studies.

A. Research Questions

The research question was formulated using the PICOC criteria, which include Population, Intervention, Comparison, Outcome, and Context. Table I shows the development of the research questions according to the PICOC framework.

TABLE I. PICOC PROTOCOL – BASED RESEACH QUESTION

Population	Intervention	Comparison	Outcomes	Context
infant, baby	audio signal processing	-	Features that are widely used for audio extraction	hospital, academic, industry

Source: Data Processed by the Author 2024, Yogyakarta

Table I demonstrates how the PICOC framework helps researchers craft specific research questions, set appropriate inclusion and exclusion criteria, and effectively guide the literature search process to organize the review on audio data augmentation.

This research aims to examine audio extraction features that are applicable to the analysis of infant cries. Drawing from relevant peer-reviewed studies, the following questions have been developed for this investigation:

- Which publishers have contributed the most to the literature on audio signal processing methods for infant cry recognition and interpretation?
- What are the most commonly used audio signal processing techniques for recognizing and interpreting infant cries?
- What are the advantages and limitations of current audio feature extraction methods in accurately classifying various types of infant cries?

B. Search Strategies

The search was carried out using specific terms or keywords in the database to identify relevant research data. To locate articles that address the research question, the keywords 'audio signal processing', 'infant crying', and 'baby crying' were utilized [30][31][32][33][34][35].

The digital libraries that will be used for this systematic literature review include IEEE Access, Science Direct, ACM Digital Library, Springer, and Sage. Scopus is also utilized because of its reputation as a comprehensive and reliable data source, often regarded as more robust than other databases.

After selecting the digital library, the next step is to design the search string. The search string used is: (infant OR baby) AND cry AND (detect* OR classificat*).

The search process involved gathering articles from the previously mentioned digital libraries, resulting a total of 2,160 articles. To focus on recent advancements, the search was limited to articles published from 2020 to 2024. References from studies that fit these criteria were then chosen for further analysis.

C. Selection Criteria and Process

After obtaining the reference list, the next step was to assess each piece of literature to determine whether it should be included for data extraction and analysis.

TABLE II. SELECTION CRITERIA

Criteria	Description
Year of Publication	2020-2024
Language	English
Study Focus	Audio Signal Processing of Infant cry
Publication	Research with verified publications
Article accessibility	Articles that are open access and available for analysis, excluding book chapters, books, irrelevant subjects
Category	Articles that clearly identify and address at least one type of bias within their content

Source: Data Processed by the Author 2024, Yogyakarta

1) Inclusion: Inclusion criteria were based on the study's objectives, requiring articles to focus on infant cry categorization techniques, maintain high research quality, and come from peer-reviewed sources. Only recent publications and key seminal studies were considered to reflect the latest advancements in the field.

2) Exclusion: Articles unrelated to audio signal processing of infant cries were excluded, as well as those with weak research designs, unrepresentative samples, or flawed data analysis. Non-peer-reviewed articles, duplicates, and works already covered in the review were also excluded to avoid redundancy.

The selection process will occur in several stages, including evaluating titles, reviewing abstracts and keywords, and assessing the content of the articles.

We initially identified 2,160 articles, which were reduced to 810 through automated publication selection tools. Title screening and abstract review further narrowed this number down to 107, and additional filtering of the full texts reduced it to 43. Ultimately, 28 articles were chosen during the final content review stage.

III. RESULT AND DISCUSSION

This chapter presents the findings and discussions derived from an extensive literature review on audio feature extraction for the analysis and interpretation of infant cries.

A. *RQ1: Which publishers have contributed the most to the literature on audio signal processing methods for infant cry recognition and interpretation?*

In the last five years, from 2020 to 2024, there has been a marked increase in research on audio signal processing, especially between 2023. Among the 28 articles reviewed, eleven were sourced from ScienceDirect, one from Sage, twelve from IEEE Access, three from ACM Digital Library, and one from Springer (see Table 3).

TABLE III. DISTRIBUTION OF ARTICLE

Year	Science Direct	IEEE Access	ACM Digital Library	Sage	Springer	Total
2020	2	1				3
2021	1	2	2			5
2022	4	2		1		7
2023	2	4	1		1	8
2024	2	3				5
Total	11	12	3	1	1	28

Source: Data Processed by the Author 2024, Yogyakarta

B. *RQ2: What are the most commonly used audio signal processing techniques for recognizing and interpreting infant cries?*

Among the 24 studies reviewed, several techniques for audio extraction are employed. Although Mel-Frequency Cepstral Coefficients (MFCCs) are the most frequently employed technique, some studies use a combination of various methods, as shown in Figure 2.

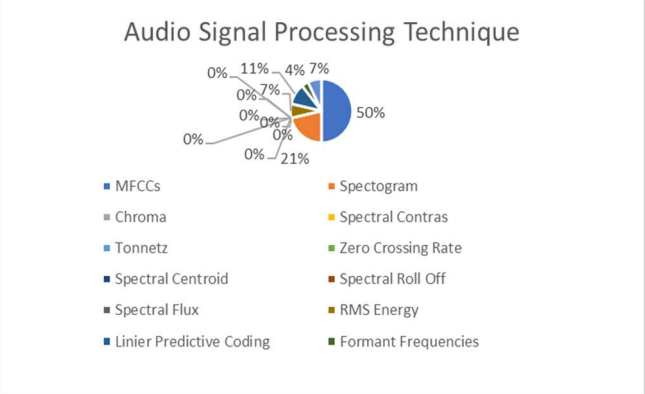


Fig. 2. Audio feature extraction for processing infant cry signals (Source: Data Processed by the Author 2024, Yogyakarta)

C. *RQ3: What are the advantages and limitations of current audio feature extraction methods in accurately classifying various types of infant cries?*

Current audio feature extraction methods for classifying baby cries have pros and cons. MFCCs, a widely used technique, are effective at modeling human hearing, making them suitable for identifying different cry patterns, as illustrated in Figure 2.

TABLE IV. REFERENCE OF THE RESEARCH QUESTION

Research Question	Description	Reference
Publisher	publishers who contribute a lot about audio signal processing research related to infant cry	[36] [37] [18] [38] [11] [13] [39] [40] [12] [41] [42] [43]
Audio Signal Processing Technique	commonly utilized audio signal processing techniques to assist in extracting infant cry audio	[36] [9] [37] [38] [11] [39] [29] [44] [45] [25] [46]
Limitation Research	limitations of current audio feature extraction methods in accurately classifying various types of infant cries	[47] [48] [28] [12] [21] [18] [38] [13] [39] [49] [29] [27] [31] [50] [44] [45] [51] [25] [46] [52] [53], [54] [26]

Source: Data Processed by the Author 2024, Yogyakarta

Table 4 explains that out of 28 articles, 14 articles are from IEEE access publishers, 11 articles focus on discussing MFCCs being a widely used audio extraction feature for infant cry and 23 articles provide an overview of the limitations contained in the research.

IV. CONCLUSION

This review finds that while audio signal processing methods for infant cry recognition have advanced, there are still notable gaps. Techniques like MFCCs and spectrograms are effective but limited by noise and the complexity of cries. MFCCs capture key cry patterns but struggle with noise and temporal relationships. Combining multiple methods and advanced models like deep learning could improve accuracy. Future research should focus on refining these techniques for better real-world application, enhancing neonatal care.

REFERENCES

[1] G. Vankudre, V. Ghulaxe, A. Dhomane, S. Badlani, and T. Rane, "A Survey on Infant Emotion Recognition through Video Clips," in Proceedings of 2nd IEEE International Conference on Computational Intelligence and Knowledge Economy, ICCIKE 2021, Institute of Electrical and Electronics Engineers Inc., Mar. 2021, pp. 296–300. doi: 10.1109/ICCIKE51210.2021.9410786.

[2] H. Alam et al., "IoT Based Smart Baby Monitoring System with Emotion Recognition Using Machine Learning," Wirel Commun Mob Comput, vol. 2023, 2023, doi: 10.1155/2023/1175450.

[3] K. S. Alishamol, T. T. Fousiya, K. J. Babu, M. Sooryadas, and L. Mary, "System for infant cry emotion recognition using DNN," in Proceedings of the 3rd International Conference on Smart Systems and Inventive Technology, ICSSIT 2020, Institute of Electrical and Electronics Engineers Inc., Aug. 2020, pp. 867–872. doi: 10.1109/ICSSIT48917.2020.9214198.

[4] A. M. Mahmoud, S. M. Swilem, A. S. Alqarni, and F. Haron, "Infant Cry Classification Using Semi-supervised K-Nearest Neighbor Approach," in Proceedings - International Conference on Developments in eSystems Engineering, DeSE, Institute of Electrical and Electronics Engineers Inc., Dec. 2020, pp. 305–310. doi: 10.1109/DeSE51703.2020.9450239.

[5] J. Li, M. Hasegawa-Johnson, and N. L. McElwain, "Analysis of acoustic and voice quality features for the classification of infant and mother vocalizations," Speech Commun, vol. 133, pp. 41–61, Oct. 2021, doi: 10.1016/j.specom.2021.07.010.

[6] C. C. Onu et al., "A cry for help: Early detection of brain injury in newborns," 2023, doi: 10.48550/arXiv.2310.08338.

[7] R. Jahangir, "CNN-SCNet: A CNN net-based deep learning framework for infant cry detection in household setting," Engineering Reports, 2023, doi: 10.1002/eng2.12786.

[8] K. Ashwini, P. M. D. R. Vincent, K. Srinivasan, and C. Y. Chang, "Deep Learning Assisted Neonatal Cry Classification via Support Vector Machine Models," Front Public Health, vol. 9, Jun. 2021, doi: 10.3389/fpubh.2021.670352.

[9] A. Abbaskhah, H. Sedighi, and H. Marvi, "Infant cry classification by MFCC feature extraction with MLP and CNN structures," Biomed Signal Process Control, vol. 86, Sep. 2023, doi: 10.1016/j.bspc.2023.105261.

[10] D. Budaghyan, C. C. Onu, A. Gorin, C. Subakan, and D. Precup, "CRYCELEB: A SPEAKER VERIFICATION DATASET BASED ON INFANT CRY SOUNDS," in ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 11966–11970. doi: 10.1109/ICASSP48485.2024.10448292.

[11] A. Gorin, C. Subakan, S. Abdoli, J. Wang, S. Latremouille, and C. Onu, "Self-Supervised Learning for Infant Cry Analysis," in ICASSPW 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing Workshops, Proceedings, Institute of Electrical and Electronics Engineers Inc., 2023. doi: 10.1109/ICASSPW59220.2023.10193421.

[12] C.-M. Chang, H. Y. Chen, H.-C. Chen, and C.-C. Lee, "Sensing with Contexts: Crying Reason Classification for Infant Care Center with Environmental Fusion."

[13] Z. Firas, A. A. Nashaat, and G. Ahmad, "Optimizing Infant Cry Recognition: A Fusion of LPC and MFCC Features in Deep Learning Models," in International Conference on Advances in Biomedical Engineering, ICABME, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 232–237. doi: 10.1109/ICABME59496.2023.10293083.

[14] N. Saude and P. A. H. Vardhini, "IoT based Smart Baby Cradle System using Raspberry Pi B+," in Proceedings of the 2020 International Conference on Smart Innovations in Design, Environment, Management, Planning and Computing, ICSIDEMPC 2020, Institute of Electrical and Electronics Engineers Inc., Oct. 2020, pp. 273–278. doi: 10.1109/ICSIDEMPC49020.2020.9299602.

[15] S. Joseph, J. Ajay Gautham, A. Kumar, and M. K. Harish Babu, "IOT Based Baby Monitoring System Smart Cradle," in 2021 7th International Conference on Advanced Computing and Communication Systems, ICACCS 2021, Institute of Electrical and Electronics Engineers Inc., Mar. 2021, pp. 748–751. doi: 10.1109/ICACCS51430.2021.9442022.

[16] A. Kumar, D. R. P. M. Vincent, K. Srinivasan, and C. Y. Chang, "Deep Convolutional Neural Network based Feature Extraction with optimized Machine Learning Classifier in Infant Cry Classification," in 2020 International Conference on Decision Aid Sciences and Application, DASA 2020, Institute of Electrical and Electronics Engineers Inc., Nov. 2020, pp. 27–32. doi: 10.1109/DASA51403.2020.9317240.

[17] S. Al-Azani, S. M. Sait, and K. A. Al-Utaibi, "A Comprehensive Literature Review on Children's Databases for Machine Learning Applications," 2022, Institute of Electrical and Electronics Engineers Inc. doi: 10.1109/ACCESS.2022.3146008.

[18] A. Abbasi, A. R. R. Javed, A. Yasin, Z. Jalil, N. Kryvinska, and U. Tariq, "A Large-Scale Benchmark Dataset for Anomaly Detection and Rare Event Classification for Audio Forensics," IEEE Access, vol. 10, pp. 38885–38894, 2022, doi: 10.1109/ACCESS.2022.3166602.

[19] IEEE Computational Intelligence Society, International Neural Network Society, Institute of Electrical and Electronics Engineers, and IEEE World Congress on Computational Intelligence (2020 : Online), 2020 International Joint Conference on Neural Networks (IJCNN) : 2020 conference proceedings.

[20] C. Ji, X. Xiao, S. Basodi, and Y. Pan, "Deep Learning for Asphyxiated Infant Cry Classification Based on Acoustic Features and Weighted Prosodic Features."

[21] V. A. Kherdekar, "Convolution Neural Network Model for Recognition of Speech for Words used in Mathematical Expression," 2021.

[22] M. Viky Aditya, A. B. Setiawan, and I. A. Rabi, "IOTA (Internet of Things and Artificial Intelegence) Journal Implementation of The Speech Recognition System Using a real time web Server Based."

[23] S. Yasin et al., "Automated Speech Recognition System to Detect Babies' Feelings through Feature Analysis," Computers, Materials and Continua, vol. 73, no. 2, pp. 4349–4367, 2022, doi: 10.32604/cmc.2022.028251.

[24] K. Zhang, H. N. Ting, and Y. M. Choo, "Baby cry recognition based on WOA-VMD and an improved Dempster–Shafer evidence theory," Comput Methods Programs Biomed, vol. 245, Mar. 2024, doi: 10.1016/j.cmpb.2024.108043.

[25] H. N. Ting, Y. M. Choo, and A. Ahmad Kamar, "Classification of asphyxia infant cry using hybrid speech features and deep learning models," Expert Syst Appl, vol. 208, Dec. 2022, doi: 10.1016/j.eswa.2022.118064.

[26] G. Coro, S. Bardelli, A. Cuttano, R. T. Scaramuzzo, and M. Ciantelli, "A self-training automatic infant-cry detector," Neural Comput Appl, vol. 35, no. 11, pp. 8543–8559, Apr. 2023, doi: 10.1007/s00521-022-08129-w.

[27] G. Zeloni and F. Pavani, "Minor second intervals: A shared signature for infant cries and sadness in music," Iperception, vol. 13, no. 2, Mar. 2022, doi: 10.1177/20416695221092471.

[28] K. Rezaee, H. G. Zadeh, L. Qi, H. Rabiee, and M. R. Khosravi, "Can You Understand Why I Am Crying? A Decision-making System for Classifying Infants' Cry Languages Based on DeepSVM Model," ACM Transactions on Asian and Low-Resource Language Information Processing, vol. 23, no. 1, Jan. 2024, doi: 10.1145/3579032.

[29] L. Klicker, A. Maliwat, J. Findura, X. Yu, M. Gromov, and J. Saniie, "InfaSafe: A Comprehensive, Non-invasive Infant Monitoring System," in IEEE International Conference on Electro Information Technology, IEEE Computer Society, 2024, pp. 291–296. doi: 10.1109/eIT60633.2024.10609853.

[30] M. T. A. Sampurna et al., "Determinants of neonatal deaths in Indonesia: A national survey data analysis of 10,838 newborns," Heliyon, vol. 9, no. 1, Jan. 2023, doi: 10.1016/j.heliyon.2023.e12980.

[31] F. Anders, M. Hlawitschka, and M. Fuchs, "Automatic classification of infant vocalization sequences with convolutional neural networks," Speech Commun, vol. 119, pp. 36–45, May 2020, doi: 10.1016/j.specom.2020.03.003.

[32] M. Charola, A. Kachhi, and H. A. Patil, "Whisper Encoder features for Infant Cry Classification," in Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH, International Speech Communication Association, 2023, pp. 1773–1777. doi: 10.21437/Interspeech.2023-1916.

[33] T. Ozseven, "A Review of Infant Cry Recognition and Classification based on Computer-Aided Diagnoses," in HORA 2022 - 4th International Congress on Human-Computer Interaction, Optimization and Robotic Applications, Proceedings, Institute of Electrical and Electronics Engineers Inc., 2022. doi: 10.1109/HORA55278.2022.9800038.

[34] A. Chaiwachiragompol and N. Suwannata, "The Study of Learning System for Infant Cry Classification Using Discrete Wavelet Transform and Extreme Machine Learning," Ingenierie des Systemes d'Information, vol. 27, no. 3, pp. 433–440, Jun. 2022, doi: 10.18280/isi.270309.

[35] A. Ekinci and E. Küçükkülahli, "Classification of Baby Cries Using Machine Learning Algorithms," 2023.

[36] K. Zaman, M. Sah, C. Direkoglu, and M. Unoki, "A Survey of Audio Classification Using Deep Learning," IEEE Access, vol. 11, pp. 106620–106649, 2023, doi: 10.1109/ACCESS.2023.3318015.

[37] X. Yao, M. Micheletti, M. Johnson, E. Thomaz, and K. de Barbaro, "INFANT CRYING DETECTION IN REAL-WORLD ENVIRONMENTS," in ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 131–135. doi: 10.1109/ICASSP43922.2022.9746096.

[38] N. Nimbarte, H. Khan, M. D. Sendre, K. Ramteke, and S. Wairagade, "New Born Baby Cry Analysis and Classification," in 2023 4th International Conference for Emerging Technology, INCET 2023, Institute of Electrical and Electronics Engineers Inc., 2023. doi: 10.1109/INCET57972.2023.10170511.

[39] H. A. Patil, A. Kachhi, and A. T. Patil, "CQT-Based Cepstral Features for Classification of Normal vs. Pathological Infant Cry," IEEE/ACM Trans Audio Speech Lang Process, 2023, doi: 10.1109/TASLP.2023.3325971.

[40] D. Budaghyan, C. C. Onu, A. Gorin, C. Subakan, and D. Precup, "CryCeleb: A Speaker Verification Dataset Based on Infant Cry Sounds," May 2023, [Online]. Available: http://arxiv.org/abs/2305.00969

[41] J. Xie, X. Long, R. A. Otte, and C. Shan, "Convolutional Neural Networks for Audio-Based Continuous Infant Cry Monitoring at Home," IEEE Sens J, vol. 21, no. 24, pp. 27710–27717, Dec. 2021, doi: 10.1109/JSEN.2021.3123906.

[42] K. Zhang, H. N. Ting, and Y. M. Choo, "Baby Cry Recognition by BCRNet Using Transfer Learning and Deep Feature Fusion," IEEE Access, vol. 11, pp. 126251–126262, 2023, doi: 10.1109/ACCESS.2023.3330789.

[43] B. N. Pradhan et al., "Baby Cry Decoder: A boon for the Hearing Impaired Caregiver," in MysuruCon 2022 - 2022 IEEE 2nd Mysore Sub Section International Conference, Institute of Electrical and Electronics Engineers Inc., 2022. doi: 10.1109/MysuruCon55714.2022.9972437.

[44] M. S. Salekin, G. Zamzmi, D. Goldgof, R. Kasturi, T. Ho, and Y. Sun, "Multimodal spatio-temporal deep learning approach for neonatal postoperative pain assessment," Comput Biol Med, vol. 129, Feb. 2021, doi: 10.1016/j.compbiomed.2020.104150.

[45] F. Salehian Matikolaie, Y. Kheddache, and C. Tadj, "Automated newborn cry diagnostic system using machine learning approach," Biomed Signal Process Control, vol. 73, Mar. 2022, doi: 10.1016/j.bspc.2021.103434.

[46] T. Ozseven, "Infant cry classification by using different deep neural network models and hand-crafted features," Biomed Signal Process Control, vol. 83, May 2023, doi: 10.1016/j.bspc.2023.104648.

[47] R. Garg, "'its Changes so Often': Parental Non-/Use of Mobile Devices while Caring for Infants and Toddlers at Home," Proc ACM Hum Comput Interact, vol. 5, no. CSCW2, Oct. 2021, doi: 10.1145/3479513.

[48] F. Anders, M. Hlawitschka, and M. Fuchs, "Comparison of artificial neural network types for infant vocalization classification," IEEE/ACM Trans Audio Speech Lang Process, vol. 29, pp. 54–67, 2021, doi: 10.1109/TASLP.2020.3037414.

[49] D. Budaghyan, C. C. Onu, A. Gorin, C. Subakan, and D. Precup, "CryCeleb: A Speaker Verification Dataset Based on Infant Cry Sounds," May 2023, [Online]. Available: http://arxiv.org/abs/2305.00969

[50] F. Salehian Matikolaie and C. Tadj, "On the use of long-term features in a newborn cry diagnostic system," Biomed Signal Process Control, vol. 59, May 2020, doi: 10.1016/j.bspc.2020.101889.

[51] T. Wang, H. Guo, Q. Zhang, and Z. Yang, "A new multilayer graph model for speech signals with graph learning," Apr. 15, 2022, Elsevier Inc. doi: 10.1016/j.dsp.2021.103360.

[52] A. Laguna et al., "Multi-modal analysis of infant cry types characterization: Acoustics, body language and brain signals," Comput Biol Med, vol. 167, Dec. 2023, doi: 10.1016/j.compbiomed.2023.107626.

[53] U. Bilotti, C. Bisogni, M. De Marsico, and S. Tramonte, "Multimodal Emotion Recognition via Convolutional Neural Networks: Comparison of different strategies on two multimodal datasets," Eng Appl Artif Intell, vol. 130, Apr. 2024, doi: 10.1016/j.engappai.2023.107708.

[54] I. J. Jung and W. H. Cho, "A novel visual representation method for multi-dimensional sound scene analysis in source localization problem," Mech Syst Signal Process, vol. 208, Feb. 2024, doi: 10.1016/j.ymssp.2023.110977.